Python Programming — Day 5 Homework

for/while loops · break/continue · Lists · Tuples · Sets

Name: ______________________________

Submit before next class

CORE TASKS — COMPLETE ALL 5

1 Number patterns

Easy

Use a for loop to print all multiples of 3 from 3 to 30 on one line each. Then on the next loop, print them in reverse order (30 down to 3).

```
Expected output (part 1):
3
6
9
...
30

Expected output (part 2):
30
27
...
3
```

Hint: for part 1 use range(3, 31, 3). For part 2 use range(30, 2, -3) — a negative step goes backwards.

2 Skip and stop

Easy

Loop through numbers 1 to 20. Use continue to skip all even numbers. Use break to stop when you reach 15. Print only the odd numbers up to 13.

```
Expected output:
1 3 5 7 9 11 13

(stops before 15, skips all evens)
```

Hint: use num % 2 != 0 to check odd. Check for break BEFORE the continue check.

3 List manager

Medium

Start with this list: [44, 12, 88, 5, 33, 88, 7]. Do all the following steps in order and print the list after each step.

```
1. Append 99
2. Insert 0 at index 2
3. Remove the first 88
4. Sort ascending
5. Pop the last item
6. Print count of 88
7. Print index of 33
```

Hint: each method modifies the list in place — print(nums) after each step to track changes.

4 Unique visitor tracker

Medium

You are given a list of website visitors (with duplicates). Use a set to find unique visitors. Then check if a specific user visited using the 'in' keyword.

```
visitors = ["alice","bob","alice","carol","
bob","dave","alice"]

1. Print total visits (len of list)
2. Print unique visitors using set
3. Print count of unique visitors
4. Check if "eve" visited → print result
5. Check if "bob" visited → print result
```

Hint: unique = set(visitors) converts to a set. Use "bob" in unique to check membership.

5 Number guessing game with while loop

Challenge

Extend the class number-guessing game using a while loop. Give the player 3 attempts. If they guess correctly, print 'You won!' and stop. If all 3 attempts are used up, reveal the secret number.

```
import random
secret = random.randint(1, 10)
attempts = 3

while attempts > 0:
guess = int(input('Guess (1-10): '))
if guess == secret:
print('You won!')
break
else:
attempts -= 1
print('Wrong!', attempts, 'attempts left')
else:
print('Game over! Number was:', secret)
```

Hint: the while...else block is a bonus Python feature — the else runs only if the loop ended without a break. Try to understand how it works by running different cases.

BONUS — OPTIONAL CHALLENGE

B Set operations – social media mutual friends

Optional

Given: `alice_friends = {"bob","carol","dave","eve"}` and `bob_friends = {"carol","frank","eve","grace"}` — find: (1) mutual friends of Alice and Bob, (2) all unique people in both friend lists combined, (3) people who are Alice's friends but NOT Bob's, (4) whether "dave" is in bob's friend list.